

A statistical mechanics framework for Bayesian deep neural networks beyond the infinite-width limit

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Despite the practical success of deep neural networks, a comprehensive theoretical framework that can predict practically relevant scores, such as the test accuracy, from knowledge of the training data is currently lacking. Huge simplifications arise in the infinite-width limit, in which the number of units N_ℓ in each hidden layer ($\ell = 1, \dots, L$, where L is the depth of the network) far exceeds the number P of training examples. This idealization, however, blatantly departs from the reality of deep learning practice. Here we use the toolset of statistical mechanics to overcome these limitations and derive an approximate partition function for fully connected deep neural architectures, which encodes information on the trained models. The computation holds in the thermodynamic limit, where both N_ℓ and P are large and their ratio $\alpha_\ell = P/N_\ell$ is finite. This advance allows us to obtain: (1) a closed formula for the generalization error associated with a regression task in a one-hidden layer network with finite α_1 ; (2) an approximate expression of the partition function for deep architectures (via an effective action that depends on a finite number of order parameters); and (3) a link between deep neural networks in the proportional asymptotic limit and Student's t -processes.

The rise of deep learning, driven by advances in computing technology and foreshadowed by decades of research, has outpaced our ability to develop a solid theoretical foundation^{1,2}. Filling the gaps in our understanding of deep learning on a fundamental level is a long-time collective effort involving several communities. Statistical physics achieved far-reaching results in this regard, and remains a wellspring of fresh perspectives and breakthroughs^{3–11}. One notable recent advance was obtained by considering the infinite-width limit, where the number of training data P is fixed and the size of the hidden layers is taken to infinity. The observation that such deep models are equivalent to Gaussian

processes (GPs)^{12–20} established a connection between deep learning and kernel methods²¹, and provided a statistical physics description of this regime^{7,22,23}.

However, there is agreement that a more complete theory should address deep learning beyond the infinite-width limit^{24–29}: in fact, realistic neural networks operate in a qualitatively different regime, where the number of training examples exceeds the width of the largest layer. Modelling the finite-width regime in the thermodynamic limit, in which the number of degrees of freedom diverges and the tools of statistical mechanics are most effective, amounts to taking the asymptotic limit

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in which both the size of the training set P and the number of units in each hidden layer N_ℓ are taken to infinity with their ratios fixed, as we consider in this work:

$$P, N_\ell \rightarrow \infty, \alpha_\ell = \frac{P}{N_\ell} \text{ finite } \forall \ell = 1, \dots, L \quad (1)$$

with L being the (finite) depth of the network (the scaling of P with the input size N_0 , which deserves special care, is discussed in ‘Constraints on the scaling...’ section in Methods). This choice guarantees that such networks work in the overparameterized regime.

Another fruitful line of research, in the direction of overcoming the limitations of the infinite-width limit, sacrifices the nonlinear nature of the network by considering a deep linear input–output mapping: even if the resulting architecture lacks the expressive power^{30–37} of the same model with non-linearities, the multilayer structure maintains the learning problem non-convex, while amenable to analytical investigation^{38,39}. Li and Sompolinsky⁵ very recently proposed a method to analytically evaluate properties of deep finite-width linear networks (for example, their generalization error) trained on a generic fixed training set; however, the more relevant case of generic nonlinear DNNs remains an open problem, despite some recent notable attempts to address it^{3,26–29,40–42}.

In statistical mechanics, the partition function is the central object encoding the properties of the system in the thermodynamic limit. In this work we address the analytical computation of the partition function of a fully connected, multilayer, nonlinear neural network, as a function of the training set in the asymptotic limit defined in equation (1). Technically, the computation amounts to integrating out an extensive number of degrees of freedom (the weights of the network), thus landing on an expression that involves only a finite number (proportional to the depth L) of integrals, to be evaluated by the saddle-point method. In the one-hidden-layer (1HL) case, the only key approximation is justified by a generalized central limit theorem due to Bardet and Surgailis⁴³ (which belongs to a class of results known as Breuer–Major theorems⁴⁴ from the seminal paper⁴⁵).

In the general case of an architecture with L hidden fully connected layers, we show that the distribution of the pre-activations at each layer ℓ is a mixture of Gaussians that depends on ℓ parameters. Notably, the backpropagating integration performed in ref. 5 is not a viable option once non-linearities are added to the model. We introduce a forward-propagating method to perform nested integrations starting from the input layer. This result depends on an assumption that is similar, at least in spirit, to the Gaussian equivalence principle employed for random and generic feature models^{46–52}.

From these developments, we are able to obtain quantitative predictions for the generalization error of the network below the interpolation threshold. Moreover, our results have an intriguing interpretation from the point of view of stochastic processes: we show that in the case of finite α_ℓ the GP arising in the infinite-width limit of Bayesian neural networks¹⁵ should be generalized to a Student’s t stochastic process⁵³.

As a first application of the theory, we establish a simple criterion (equivalent to the one found in the linear case^{5,42} and in finite- P perturbation theory^{40,54}) to predict whether it is convenient, in terms of generalization performance, to employ a finite-width deep neural network over its infinite-width version.

Problem setting

We consider a supervised learning problem with training set $\mathcal{T}_P = \{\mathbf{x}^\mu, \mathbf{y}^\mu\}_{\mu=1}^P$, where each $\mathbf{x}^\mu \in \mathbb{R}^{N_0}$ and the corresponding labels $\mathbf{y}^\mu \in \mathbb{R}$. The architecture is a deep neural network $f_{\text{DNN}}(\mathbf{x})$ with $(L-1)$ fully connected hidden layers and a final linear readout layer as defined in equation (22). We analyse regression problems with a quadratic loss function:

$$\mathcal{L} = \frac{1}{2} \sum_{\mu=1}^P [y^\mu - f_{\text{DNN}}(\mathbf{x}^\mu)]^2 + \mathcal{L}_{\text{reg}}, \quad (2)$$

$$\mathcal{L}_{\text{reg}} = \frac{\lambda_L}{2\beta} \sum_{i_L=1}^{N_L} v_{i_L}^2 + \frac{1}{2\beta} \sum_{\ell=0}^{L-1} \lambda^{(\ell)} \|W^{(\ell)}\|^2, \quad (3)$$

where L^2 regularizations have been added for each layer to the loss function, $\|\cdot\|$ is the standard Frobenius norm defined for the weights matrices $W^{(\ell)}$, and $\beta = 1/T$ is the inverse temperature parameter.

As a standard practice in statistical mechanics of deep learning, we define the partition function of the problem as:

$$Z = \int \mathcal{D}\theta e^{-\beta \mathcal{L}(\theta)}. \quad (4)$$

where the symbol $\int \mathcal{D}\theta$ indicates the collective integration over the weights of the network, $\theta = \{W^{(\ell)}, v\}$. This choice enforces minimization of the training error for $\beta \rightarrow \infty$. We notice that scaling $\mathcal{L}_{\text{reg}}$ by $1/\beta$ has a natural Bayesian learning interpretation: the Gibbs probability $P_\beta(\theta) = Z^{-1} e^{-\beta \mathcal{L}(\theta)}$ associated with the partition function in equation (4) is the posterior distribution of the weights after training, whereas the Gaussian regularization is a prior equivalent to assuming that weights at initialization have been drawn from a Gaussian distribution.

In this framework, the average test error over a new (unseen) example $(\mathbf{x}^0, y^0)$ is given by:

$$\langle \epsilon_g(\mathbf{x}^0, y^0) \rangle = \int \mathcal{D}\theta [y^0 - f_{\text{DNN}}(\mathbf{x}^0)]^2 \frac{e^{-\beta \mathcal{L}(\theta)}}{Z}. \quad (5)$$

Results

Asymptotic Bayesian effective action for 1HL NNs. In the case of 1HL architectures, we are able to reduce the partition function (equation (4)) to the following two-variables integral in the thermodynamic limit described in equation (1):

$$Z = \int dQ \int d\bar{Q} \exp \left[-\frac{N_1}{2} S(Q, \bar{Q}) \right] \quad (6)$$

where we have defined an effective action S given by:

$$S = -Q\bar{Q} + \log(1 + Q) + \frac{\alpha_1}{P} \text{Tr} \log \beta \left[\frac{1}{\beta} + \frac{QK}{\lambda_1} \right] + \frac{\alpha_1}{P} \mathbf{y}^\top \left[\frac{1}{\beta} + \frac{QK}{\lambda_1} \right]^{-1} \mathbf{y} \quad (7)$$

and we have introduced a vectorial notation for the output $\mathbf{y}^\top = (y^1, y^2, \dots, y^P)$. The $P \times P$, input-dependent kernel K/λ_1 is the neural network Gaussian process (NNGP) kernel¹⁵ arising in the infinite-width limit and its precise definition in terms of the input covariance matrix (rescaled by the Gaussian prior of the first layer λ_0) $C_{\mu\nu} = \mathbf{x}^\mu \cdot \mathbf{x}^\nu / (\lambda_0 N_0)$ is given in Methods, equation (44). Also note that equation (7) holds for zero-mean activation functions, that is, functions whose average over a centred Gaussian is zero (see equation (43)); an effective action for the generic finite-mean case is reported in Supplementary Section 4, and that the derivation goes through for many reasonable non-linearities and input data distributions, at least in the regime $P = O(N_0)$ (we discuss this key technical point in Methods). This is the first main result of our work, and we conjecture it is exact because the only key Gaussian approximation that we perform is justified by the extension of the Breuer–Major theorem in ref. 43, as argued in Methods.

In the Supplementary Information we obtain a number of additional results that were not entered here due to space limitations: (1) a rederivation of the effective action in the case of linear activation function, valid at fixed P, N_1, N_0 , together with a comparison with the results given in refs. 5,55; (2) a specific derivation of the effective action for quadratic activation function, which makes no use of the Breuer–Major

theorem; (3) the generalization of the effective action in equation (7) to the case of multiple (but finite) outputs.

We can now solve equation (7) using the saddle-point method, as $N_1 \rightarrow \infty$, which amounts to finding the solutions Q^* , $\bar{Q}^*$ of the system of equations $\partial_Q S = 0$, $\partial_{\bar{Q}} S = 0$ (the infinite-width limit is reobtained for $\alpha_1 \rightarrow 0$ and corresponds to the particular solution $Q^*, \bar{Q}^* = 1$). In the zero-temperature limit we can find the analytical solution of the saddle-point equations (see Methods). A straightforward computation shows that the generalization error is given in terms of the usual bias-variance decomposition:

$$\begin{aligned} \langle \epsilon_g(\mathbf{x}^0, y^0) \rangle &= (y^0 - \Gamma_1)^2 + \sigma_1^2, \\ \Gamma_1 &= \sum_{\mu, \nu} \kappa_\mu(\mathbf{x}^0) K_{\mu\nu}^{-1} y_\nu, \\ \sigma_1^2 &= \frac{Q^*}{\lambda_1} \left[\kappa_0(\mathbf{x}^0) - \sum_{\mu, \nu} \kappa_\mu(\mathbf{x}^0) K_{\mu\nu}^{-1} \kappa_\nu(\mathbf{x}^0) \right], \end{aligned} \quad (8)$$

where $\kappa_\mu(\mathbf{x}^0)$, $\kappa_0(\mathbf{x}^0)$ can be computed from the functional definition of the NNGP kernel using the new unseen input $\mathbf{x}^0$, as shown in Methods.

We can directly employ equation (8) to obtain testable predictions for the generalization error of finite-width IHL architectures trained in the Bayesian learning setting, as we do in Fig. 1a for two specific regression tasks defined on the CIFAR10 and MNIST datasets (refer to ‘Numerical experiments’ section of Methods, as well as the Supplementary Information, for further details). We find that the generalization curves for the two regression tasks are monotonically increasing (decreasing) as a function of N_1 depending on the fact that the observable $y^\top (K/\lambda_1)^{-1} y/P$ is smaller (larger) than 1. The importance of this quantity in controlling the generalization performance has been already noted in linear networks^{5,42} as well as in direct perturbation theory at finite P for nonlinear networks^{40,54}.

We also point out two semi-quantitative predictions for the general behaviour of the generalization error, just by looking at the dependence of equation (8) on the size of the hidden layer N_1 and on the Gaussian prior of the last layer λ_1 . At $T = 0$, the bias is constant as a function of N_1 (as explicitly observed also in the linear case in ref. 5) and λ_1 . By contrast, the variance depends on N_1 and decreases as $1/\sqrt{\lambda_1}$ in the large- λ_1 limit. These observations lead to the following two testable predictions: (1) increasing the magnitude of the Gaussian prior λ_1 should systematically improve the generalization performance at any N_1 ; (2) for large λ_1 the dependence on N_1 of the generalization error should disappear (see also the numerical experiments performed in Fig. 1b).

Link between Student’s t -processes and shallow neural networks.

In obtaining the results reported in the previous section, our theory can be formulated as a statement on the probability distribution of the output variables

$$s^\mu \equiv \frac{1}{\sqrt{N_1}} \sum_{i=1}^{N_1} v_i \sigma(h_i^\mu), \quad (9)$$

where $\mathbf{h} \approx \mathcal{N}(\mathbf{0}, \mathbf{C} \otimes \mathbb{1}_{N_1})$, $\mathbf{v} \approx \mathcal{N}(\mathbf{0}, \lambda_1^{-1} \mathbb{1}_{N_1})$. Proceeding as in the derivation of the partition function presented in Methods, the probability density function of these variables can be written as a reweighted Fourier transform,

$$P(s|\mathcal{T}_P) = \frac{e^{-\frac{\beta}{2} \sum_\mu (y^\mu - s^\mu)^2}}{Z} \int \prod_\mu \frac{d\tilde{s}^\mu}{2\pi} e^{i\tilde{s}^\mu s^\mu} \Xi(\tilde{s}), \quad (10)$$

of the function

$$\Xi(\tilde{s}) = \left(1 + \frac{1}{\lambda_1 N_1} \sum_{\mu, \nu} \tilde{s}^\mu K_{\mu\nu}(C) \tilde{s}^\nu \right)^{-\frac{N_1}{2}}. \quad (11)$$

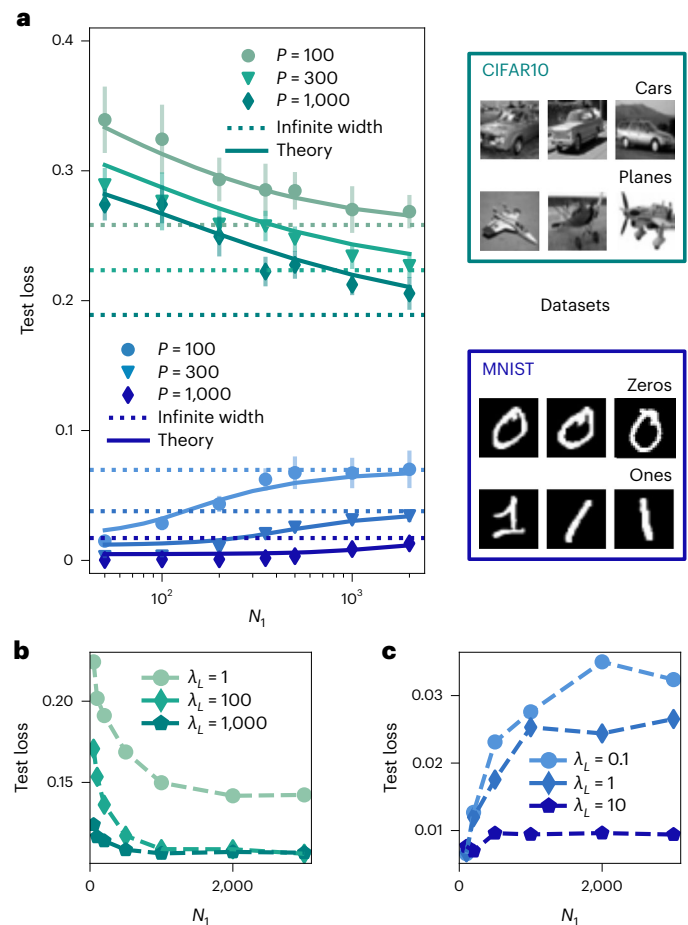


Fig. 1 | Learning curves of IHL networks. **a**, Learning curves of IHL architectures with Erf activation (trained with a discretized Langevin dynamics; see Methods for further details) as a function of the hidden layer size N_1 for two regression tasks on the CIFAR10 (above) and MNIST (below) datasets. Zero/one labels have been chosen in both cases, and the images of the CIFAR10 dataset have been greyscaled and downsampled to $N_0 = 28 \times 28$. The experimental test loss at different values of the training set size P (points with error bars indicating one s.d.) are compared with the theory computed from equation (8) (solid lines). The bar centres are computed as the average over an ensemble of $S = 450$ equilibrium configurations. Samples are taken every 10^4 Langevin steps (after thermalization of the dynamics). **b, c**, Experimental learning curves as a function of N_1 for increasing values of the Gaussian prior of the last layer λ_1 . Error bars are within points, and dashed lines serve to guide the eye. The nets are trained on $P = 3,000$ examples from the CIFAR10 dataset (**b**) and $P = 500$ examples from MNIST (**c**). Two qualitative predictions of the theory at zero temperature are checked: (1) the generalization loss should decrease for any N_1 when λ_1 grows; (2) the dependence of the learning curves on N_1 disappears in the large- λ_1 limit, since the bias is constant (see also main text).

It is straightforward to notice that as long as $N_1 \rightarrow \infty$ and $N_1 \gg P$, the dependence on N_1 disappears and we get:

$$\Xi(\tilde{s}) \rightarrow e^{-\frac{1}{2\lambda_1} \sum_{\mu, \nu} \tilde{s}^\mu K_{\mu\nu}(C) \tilde{s}^\nu}. \quad (12)$$

This quantity has a very natural interpretation in view of the NNGP literature. Indeed, for large N_1 and finite P , the variables in equation (9) are jointly multivariate Gaussian-distributed according to the central limit theorem, as noted, for example, in ref. 15: this limit corresponds indeed to the right hand side of equation (12), and is the cornerstone of the mapping of an infinite-width Bayesian neural network to a GP. However, this is no longer the case when P is comparable with N_1

(equation (11)), which is derived exploiting the Gaussian equivalence on the basis of the Breuer–Major theorem in the proportional asymptotic limit $P/N_1 \approx O(1)$, suggesting that the variables s^μ are distributed according to a multivariate Student's t -distribution^{53,56–58}.

The need of considering Student's t -processes as a generalization of NNGPs has been noted already in the case of different priors on the distribution of the last layer's weights⁵⁹. Non-Gaussianity of the posterior in a form similar to that of equation (11) has appeared also in refs. 60–62. The reason why this kind of process arises in the case we are considering here can be understood with an heuristic argument: when N_1 and P are of the same order, we cannot take the limit $N_1 \rightarrow \infty$ before $P \rightarrow \infty$, and so we need to use the empirical covariance of the output variables s^μ instead of their true one in estimating their probability distribution. A more precise characterization of these neural network Student's t -processes (NNTPs) and the regime where they arise represent interesting topics for future work.

Asymptotic Bayesian effective action for DNNs. In the generic case of a deep fully connected architecture with a finite number of layers L and zero-mean activation function, we express the partition function in terms of a $2L$ -dimensional integral (see Methods):

$$Z_{\text{DNN}} = \int \prod_{\ell=1}^L dQ_\ell d\bar{Q}_\ell e^{-\frac{N_\ell}{2} S_{\text{DNN}}(\{Q_\ell, \bar{Q}_\ell\})}, \quad (13)$$

where the effective action is given by:

$$\begin{aligned} S_{\text{DNN}} = & \sum_{\ell=1}^L \frac{\alpha_\ell}{\alpha_\ell} [-Q_\ell \bar{Q}_\ell + \log(1 + Q_\ell)] \\ & + \frac{\alpha_\ell}{P} \text{Tr} \log \beta \left(\frac{1}{\beta} + K_L^{(R)}(\{\bar{Q}_\ell\}) \right) \\ & + \frac{\alpha_\ell}{P} y^T \left(\frac{1}{\beta} + K_L^{(R)}(\{\bar{Q}_\ell\}) \right)^{-1} y \end{aligned} \quad (14)$$

and we have introduced a renormalized kernel $K^{(R)}$ that generalizes the recurrence relation for the L -layer NNGP kernel as:

$$K_\ell^{(R)}(\{\bar{Q}_\ell\}) = \bar{Q}_\ell / \lambda_\ell K_{\ell-1}^{(R)}(\{\bar{Q}_\ell\}), \quad K_0^{(R)} = C, \quad (15)$$

where C is the covariance matrix of the inputs defined above and we stress that each $K_\ell^{(R)}$ depends on the variables $\bar{Q}_1, \dots, \bar{Q}_{\ell-1}$ only. For completeness, we notice that the recurrence relation for the infinite-width kernel K_L is given by equation (15) with $\bar{Q}_\ell = 1 \forall \ell = 1, \dots, L$.

This action shares the same structure as the one found above for the special case of IHL, with the difference that for L hidden layers, the recursive nature of the derivation introduces additional order parameters that are nested in the definition of the kernel K_ℓ . Furthermore, as our derivation applies to layers of arbitrary size N_ℓ , the action also depends on the aspect ratios α_ℓ . In the Supplementary Information we derive a series of additional results: (1) we generalize this effective action for finite-mean activation functions; (2) we show how to recover the linear case in the isotropic limit $\alpha_\ell = \alpha \forall \ell = 1, \dots, L$; and (3) using (1), we demonstrate how to correct the heuristic theory for the ReLU activation presented in ref. 5.

The computation of the generalization error over a new example $(\mathbf{x}^0, y^0)$ gives:

$$\langle \epsilon_g(\mathbf{x}^0, y^0) \rangle = (y^0 - \Gamma_L)^2 + \sigma_L^2 \quad (16)$$

where

$$\begin{aligned} \Gamma_L = & \sum_{\mu\nu} \kappa_{L\mu}^{(R)} \left(\frac{1}{\beta} + K_L^{(R)}(\{\bar{Q}_\ell\}) \right)^{-1}_{\mu\nu} y_\nu, \\ \sigma_L^2 = & \kappa_{L0}^{(R)} - \sum_{\mu\nu} \kappa_{L\mu}^{(R)} \left(\frac{1}{\beta} + K_L^{(R)}(\{\bar{Q}_\ell\}) \right)^{-1}_{\mu\nu} \kappa_{L\nu}^{(R)} \end{aligned} \quad (17)$$

and $\kappa_{L\mu}^{(R)}, \kappa_{L0}^{(R)}$ are recursive kernels computed from the recurrence given in equation (15) using the input $\mathbf{x}^0$ in the initial conditions.

Note that also in this case we can perform the same scaling analysis of the dependence of the generalization error on the Gaussian prior in the last layer λ_L (in the zero-temperature limit). It turns out that the bias does not depend on it, whereas the variance σ_L^2 approaches zero as $1/\sqrt{\lambda_L}$ as λ_L is taken to infinity. This means that also in the case of finite depth $L > 1$, training at large values of the Gaussian prior of the last layer should improve generalization at any aspect ratio of the network. We confirm this general observation with numerical experiments in Fig. 2c,f. However, unlike the IHL case, we observe that the bias does depend on the aspect ratio even in the zero-temperature limit and we cannot expect anymore that the dependence on the aspect ratios of the networks α_ℓ disappears in the $\lambda_L \rightarrow \infty$ limit.

We can obtain another prediction of the theory at L layers (which again confirms previous results on linear networks and perturbative calculations for nonlinear networks^{5,40,42,54}) by considering the effective action for ReLU activation. A straightforward Taylor expansion around the infinite-width limit $\alpha_\ell = \alpha \forall \ell = 1, \dots, L$ shows that the first correction to the test loss $\Delta \epsilon_g$ is proportional to:

$$\Delta \epsilon_g \propto \alpha \left(\frac{1}{P} y^T K_L^{-1} y - 1 \right). \quad (18)$$

where K_L is the solution of recurrence in equation (15) for $\bar{Q}_\ell = 1 \forall \ell = 1, \dots, L$ and ReLU activation. This means that there exists a simple scalar observable that determines whether the finite-width deep neural network will outperform its infinite-width counterpart that generalizes the one found at IHL:

$$s_L = \frac{1}{P} y^T K_L^{-1} y. \quad (19)$$

In particular, we expect the finite-width network to outperform its infinite-width counterpart whenever $s_L < 1$. In Fig. 2a,c we check this prediction for deep architectures with ReLU activation on the same regression tasks employed in the IHL case. Notice that s_L quickly diverges to infinity as the number of hidden layers L grows. The reason for this is simply that the ReLU NNGP kernel K_L develops at least one zero eigenvalue as $L \rightarrow \infty$. This ultimately occurs because each element of the matrix K_L converges to zero as L grows (see Fig. 2b,c), as one can easily check by looking at the explicit recurrence relation for the NNGP ReLU kernel^{15,63}. We note that this singularity can be equivalently thought as the fixed point of the discrete dynamical map defined by the recurrence relation for the NNGP kernel and therefore it might be worth investigating the relation between the generalization performance in our asymptotic limit and the line of work on the edge of chaos in random neural networks^{64,65}.

Equation (18) provides an additional link with Student's t inference. In fact, the same criterion has been found by Tracey and Wolpert⁶⁶ in the study of Bayesian optimization with Student's t -processes. The authors show that the value of s_L determines whether the Student's t -process they consider has a larger/smaller variance than the corresponding GP with the same kernel.

Discussion

In our work we have described a strategy to investigate the statistical mechanics of deep neural networks beyond the infinite-width limit, that is, in the finite asymptotic regime $P, N_\ell \rightarrow \infty$ at $\alpha_\ell = P/N_\ell > 0$ as opposed to the infinite-width $\alpha_\ell = 0$. In the IHL case, we conjecture that our evaluation is exact in the above thermodynamic limit. As such, we do not expect any further corrections to the result, at least in the asymptotic regime. In particular, we have found a closed expression for the generalization error that in principle provides a Bayesian estimator of the generalization capabilities of fully connected architectures for

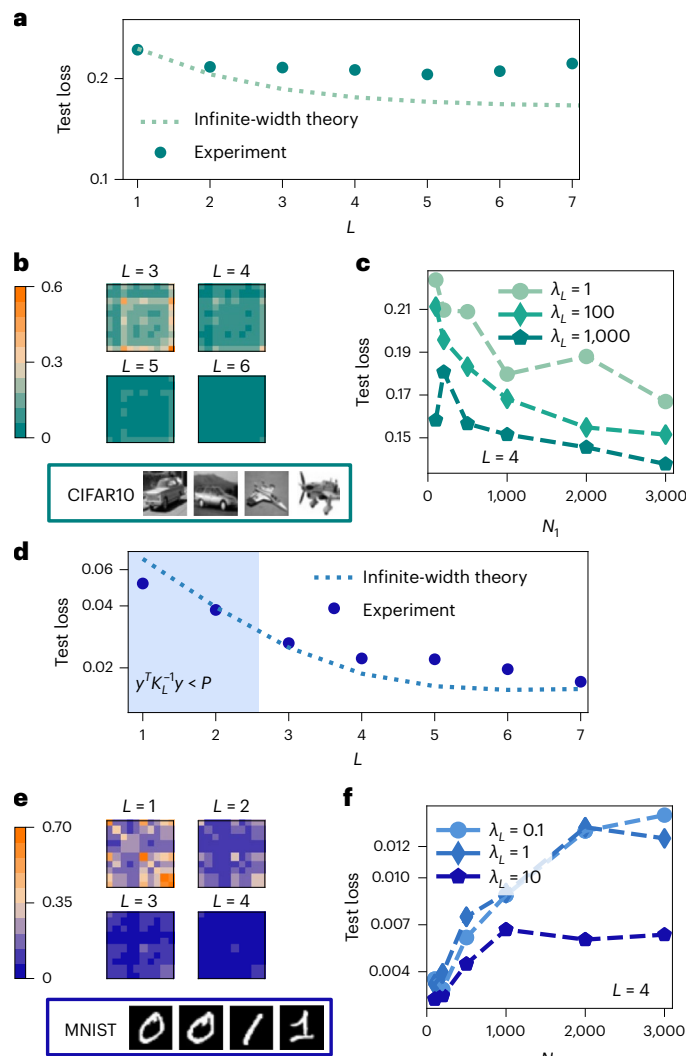


Fig. 2 | Experiments with deep networks $L > 1$. **a, d**, Test loss of a LHL neural network with ReLU activation, as a function of the depth L , for $P = 100$ (**a** shows results for CIFAR10, **d** for MNIST). The net is trained on a regression task in the small α regime ($\alpha = 0.1$), close to the infinite-width limit. The finite-width network can outperform the infinite-width prediction only when $s_L < 1$ (shaded area), that is, only for the MNIST task and for depth $L < 3$. **b, e**, Visualization of the entries of the infinite-width NNGP kernel at different layers of the network (CIFAR10 in **b**, MNIST in **e**). The ReLU NNGP kernel converges to zero after repeated iterations. This generates almost vanishing eigenvalues that makes s_L eventually always larger than one. **c, f**, Test loss of a 4HL network trained on $P = 1,000$ examples with different regularization strengths (with $N_L = N = 1,000$; CIFAR10 in **c**, MNIST in **f**). Although increasing the magnitude of the Gaussian prior of the last layer still improves generalization for all N , it is not clear anymore (as it was for 1HL networks) that the curve at large λ_L is a constant as a function of N . The dashed line is shown to guide the eye. In all panels, error bars lie within points.

any given empirical dataset, provided that the chosen architecture is capable of perfectly fitting the training set.

For the case of finite-depth $L > 1$ networks, it should be possible, at least in principle, to systematically take into account non-Gaussian corrections to the saddle-point action to check whether these are relevant for the theory at finite width, as the assumptions we made in deriving the results are clear⁶⁷ (see also Methods).

From the mathematical perspective, we find the link with Student's t -processes very promising. The precise characterization of this mapping and its limits of validity represent a research line for future investigation.

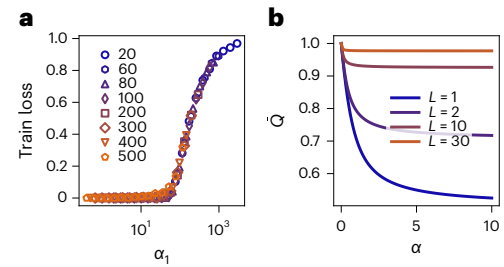


Fig. 3 | Universality behaviour of random data and order parameter as a function of depth L . **a**, Training loss of different 1HL architectures trained on a completely random task (that is, both the inputs $\mathbf{x} \in \mathbb{R}^{N_0}$ with $N_0 = 50$ and the scalar outputs y are independent and identically distributed random variables sampled from a normal distribution with zero mean and unit variance) as a function of α_1 . Error bars lie within points. At the moment we cannot capture this universal phenomenon with our theory, which only describes the overparameterized limit in which the training error is exactly zero. **b**, The numerical evaluation of the solution $\bar{Q}$ is shown in the case of ReLU activation function and isotropic network $\alpha_\ell = \alpha \forall \ell$, for different depths L . As L grows ($L = 30$), the parameter $\bar{Q}$ quickly approaches 1 for all α , suggesting that also DNNs in the asymptotic regime converge to a kernel limit in the sequential limit where the depth L is taken to infinity after P, N .

Notably, our theory predicts that a kernel limit should also appear in the asymptotic regime as the depth L approaches infinity. This could be checked, for instance, considering the isotropic limit $\alpha_\ell = \alpha \forall \ell$ and ReLU activation. Here one can numerically solve the saddle-point equation for $\bar{Q}$ at large L and verify that $\bar{Q} \rightarrow 1$ for all α , as shown in Fig. 3b. As such, also in this limit we expect an equivalence with a kernel theory with kernel given by $K_{\infty}(\mathbf{C})$. Note that from our framework it is clear that we are taking the depth L infinity only after P, N . As such we are not making claims about the challenging simultaneous limit $L, N \rightarrow \infty$ at fixed L/N , as done, for instance, in refs. 27,28,55,67.

It is fair to stress that our theory only describes the equilibrium regime of zero train loss, and thus our analysis should not apply in the regime $P/N_1 \gg 1$. However, numerical simulations performed with 1HL architectures of varying width and random training labels show that the train loss follows a universal behaviour^{42,68} with regards to α_1 (see Fig. 3a), including regimes in which the DNN is incapable of perfectly fitting the data. It would be desirable to develop a theory that also captures this phase.

Another interesting aspect to understand is the degree to which this mean-field static analysis can be extended beyond equilibrium to assess the full training dynamics; such a theory would indeed make it possible to investigate the performance of the many (often heuristic) learning algorithms employed to train deep neural networks.

We conclude by pointing out that it would be interesting to compare our theory at fixed data with the data-averaged cases studied in refs. 42,69, and to extend our results to convolutional layers, as done in the infinite-width case in ref. 17.

Methods

Setting of the learning problem and notation

We consider deep neural networks $f_{\text{DNN}}(\mathbf{x})$ with L fully connected hidden layers, where the pre-activations of each layer $h_{i_\ell}^{(\ell)}$ ($i_\ell = 1, \dots, N_\ell$; $\ell = 1, \dots, L$) are given recursively as a nonlinear function of the pre-activations at the previous layer $h_{i_{\ell-1}}^{(\ell-1)}$ ($i_{\ell-1} = 1, \dots, N_{\ell-1}$):

$$h_{i_\ell}^{(\ell)} = \frac{1}{\sqrt{N_{\ell-1}}} \sum_{i_{\ell-1}=1}^{N_{\ell-1}} w_{i_\ell i_{\ell-1}}^{(\ell)} \sigma(h_{i_{\ell-1}}^{(\ell-1)}) + b_{i_\ell}^{(\ell)}, \quad (20)$$

$$h_{i_1}^{(1)} = \frac{1}{\sqrt{N_0}} \sum_{i_0=1}^{N_0} w_{i_1 i_0}^{(1)} x_{i_0} + b_{i_1}^{(1)} \quad (21)$$

where $W^{(\ell)}$ and $b^{(\ell)}$ are the weights and biases of the ℓ th layer, respectively, whereas the input layer has dimension N_0 (the input data dimension); σ is a nonlinear activation function and it is common to each layer. We add one last readout layer and we define the function implemented by the deep neural network as:

$$f_{\text{DNN}}(\mathbf{x}) = \frac{1}{\sqrt{N_L}} \sum_{i=1}^{N_L} v_i \sigma \left[h_{i_L}^{(L)}(\mathbf{x}) \right], \quad (22)$$

where $\mathbf{v}$ is the vector of weights of the last layer.

The average training error at a given inverse temperature β is given by:

$$\langle \epsilon_t \rangle = \frac{1}{P} \int \mathcal{D}\theta [\mathcal{L}(\theta) - \mathcal{L}_{\text{reg}}(\theta)] \frac{e^{-\beta \mathcal{L}(\theta)}}{Z}, \quad (23)$$

This quantity, together with the test error defined in equation (5), represent two special observables, but more in general, for an arbitrary observable O we have:

$$\langle O \rangle = \int \mathcal{D}\theta O(\theta) \frac{e^{-\beta \mathcal{L}(\theta)}}{Z}. \quad (24)$$

The Breuer–Major theorem as a justification for the Gaussian equivalence in shallow networks

The Breuer–Major theorem and its extensions deal with the following sequence of random variables:

$$S_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N c_i F(x_i) \quad N \geq 1. \quad (25)$$

Clearly, if the distribution of the vector $\mathbf{x} = (x_1, \dots, x_N)$ is factorized over its coordinates (that is, $p(\mathbf{x}) = \prod_i p(x_i)$ and $F(x) = x$), the random variable $S = \lim_{N \rightarrow \infty} S_N$ is normal-distributed as long as the mean $\mathbb{E}(x_i) = 0$, the variance $\mathbb{E}(x_i^2)$ is finite, and the values of c_i satisfy the so-called Lindeberg's condition. This is also true whenever F is a well-behaved non-linearity.

The Breuer–Major theorem essentially extends this result to generic GPs, providing sufficient conditions on the covariance matrix of the GP, and on the non-linearity F that guarantee convergence of S_N to the normal distribution. We report here the modern statement of the theorem given in ref. 44.

We first consider a stationary (unidimensional) GP $x = (x_k)_{k \in \mathbb{Z}}$. Stationarity—which is not essential and will be replaced by a weaker condition in the following—requires that the covariance of the process $C_{ij} = \mathbb{E}(x_i x_j)$ is a function of the difference $i - j$, that is, $C_{ij} = C(i - j)$. The only technical condition to be imposed on the nonlinear function F is to have well-defined Hermite rank R . The Hermite rank is the smallest positive integer that appears in the decomposition of F over the Hermite polynomials:

$$F(x) = \sum_{k=R}^{\infty} f_k \text{He}_k(x), \quad (26)$$

where $\text{He}_k(x)$ is the k th Hermite polynomial and f_k the coefficient of the expansion. For many reasonable activation functions F , $R = 1$.

Theorem 1. This section is based on the Breuer–Major theorems from ref. 45. Let $x = (x_k)_{k \in \mathbb{Z}}$ be a stationary unidimensional GP with covariance $C(i - j)$. Let $\mathbb{E}[F(x_1)] = 0$ and $\mathbb{E}[F^2(x_1)] < \infty$ and assume that the function F has Hermite rank $R \geq 1$. Suppose that:

$$\sum_{j \in \mathbb{Z}} |C_{ij}|^R < \infty. \quad (27)$$

Then $\sigma^2 := \mathbb{E}[F(x_1)^2] + 2 \sum_{j=1}^{\infty} \mathbb{E}[F(x_1)F(x_j)]$ is finite. Moreover, one has that the sequence of random variables

$$S_N = \frac{1}{\sqrt{N}} \sum_{i=1}^N F(x_i) \quad N \geq 1 \quad (28)$$

converges in distribution to $\mathcal{N}(0, \sigma^2)$, that is, to a Gaussian distribution with zero mean and variance σ^2 .

For the scope of this work we will need a slightly stronger statement than the one just mentioned because: (1) in our calculation the covariance will not be stationary, and (2) we will need to consider a more general sequence of nonlinear functions $c_i F(x_i)$, such that each term of the sum (equation (28)) can be weighted by a factor $c_i \neq 1$.

It has been already shown in ref. 45 that the hypothesis of stationarity can be weakened and replaced with a requirement of uniform convergence of the elements of the covariance, namely:

$$\sum_{j \in \mathbb{Z}} |C_{ij}|^R < B_0 \quad \forall i \in \mathbb{Z}, \quad (29)$$

where B_0 is a positive finite constant.

Extensions (1) and (2) above have been addressed more recently by Bardet and Surgailis⁴³, as we report in the following. Let $\mathbf{x}^N$ be an N -dimensional Gaussian vector, such that $\mathbb{E}[x_i^N] = 0$, $\mathbb{E}[(x_i^N)^2] = 1$. Now define $C_{ij}^N = \mathbb{E}[x_i^N x_j^N]$. For a given integer $m \geq 1$, assume

$$\sup_{N \geq 1} \max_{1 \leq i, j \leq N} |C_{ij}^N|^m < \infty, \quad (30)$$

$$\sup_{N \geq 1} \frac{1}{N} \sum_{1 \leq i, j \leq N} |C_{ij}^N|^m \xrightarrow{K \rightarrow \infty} 0. \quad (31)$$

$|i - j| > K$

Take also $\mathbb{L}_0^2(x) = \{f : \mathbb{E}f(x) = 0, \mathbb{E}f^2(x) < \infty\}$ where x is a standard normal variable. We then continue in the next section.

Theorem 2. This section is based on the theorems of section ‘(1.ii)’ in the work by Bardet and Surgailis⁴³. Assume equations (30), (31) and let $f_i^N \in \mathbb{L}_0^2(x)$ ($N \geq 1, 1 \leq i \leq N$) be a sequence of functions all having Hermite rank m at least one. Assume that there exists a $\mathbb{L}_0^2(x)$ -valued continuous function ϕ_τ , $\tau \in [0, 1]$, such that

$$\sup_{\tau \in (0, 1]} \mathbb{E} \left[f_{[\tau N]}^N(x) - \phi_\tau(x) \right]^2 \xrightarrow{N \rightarrow \infty} 0. \quad (32)$$

Moreover, let

$$(\sigma^N)^2 = \mathbb{E} \left[\frac{1}{\sqrt{N}} \sum_{i=1}^N f_i^N(x_i^N) \right]^2 \xrightarrow{N \rightarrow \infty} \sigma^2, \quad (33)$$

where $\sigma^2 > 0$. Then

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N f_i^N(x_i^N) \xrightarrow[N \rightarrow \infty]{d} \mathcal{N}(0, \sigma^2), \quad (34)$$

where $\xrightarrow[N \rightarrow \infty]{d}$ denotes convergence in distribution.

The hypotheses of this theorem should be taken as conditions on the activation function σ , on the rescaled input covariance matrix $C_{\mu\nu}$ and on the dominant configurations $\bar{s}$ in the Fourier integral (equation (10)) to justify our Gaussian ansatz (see equation (41)).

Sketch of the calculation of the effective action in the Bayesian set-up for 1HL fully connected neural networks

We now discuss the salient aspects of the calculation. The starting point is the following partition function:

$$Z = \int \prod_{i_1}^{N_1} dv_{i_1} \prod_{i_1, i_0}^{N_1, N_0} dw_{i_1 i_0} \exp \left\{ -\frac{\lambda_1}{2} \sum_{i_1} v_{i_1}^2 - \frac{\lambda_0}{2} \|w\|^2 - \frac{\beta}{2} \left[y^\mu - \frac{1}{\sqrt{N_1}} \sum_{i_1} v_{i_1} \sigma \left(\sum_{i_0} \frac{w_{i_1 i_0} x_{i_0}^\mu}{\sqrt{N_0}} \right) \right]^2 \right\}. \quad (35)$$

where $w = W^{(1)}$ and we took $b^{(1)} = 0$ without loss of generality. (One can map a system with non-zero biases in a zero-bias system by increasing both the dimensions of the input and of the activations at each layer by one. The original biases are then trivially mapped in the extra weights of the augmented system). The first step is to decouple the weights of the different layers in the loss function. This can be done including standard identities built over two families of Dirac deltas, one for the pre-activations of the hidden layer and one for the output of the network:

$$1 = \int \prod_{\mu}^P ds^{\mu} \delta \left[s^{\mu} - \frac{1}{\sqrt{N_1}} \sum_{i_1} v_{i_1} \sigma(h_{i_1}^{\mu}) \right], \quad (36)$$

$$1 = \int \prod_{\mu}^P \prod_{i_1}^{N_1} dh_{i_1}^{\mu} \delta \left(h_{i_1}^{\mu} - \frac{1}{\sqrt{N_0}} \sum_{i_0} w_{i_1 i_0} x_{i_0}^{\mu} \right). \quad (37)$$

By using a standard Fourier representation of these deltas, which introduces the conjugate variables $\bar{h}_{i_1}^{\mu}$ and $\bar{s}^{\mu}$, we can perform the Gaussian integrals on the internal and external weights:

$$Z = \int \prod_{\mu}^P \frac{ds^{\mu} d\bar{s}^{\mu}}{2\pi} e^{-\frac{\beta}{2} \sum_{\mu} (y^{\mu} - s^{\mu})^2 + i \sum_{\mu} s^{\mu} \bar{s}^{\mu}} \times \left\{ \int \prod_{\mu}^P \frac{dh_{i_1}^{\mu} d\bar{h}_{i_1}^{\mu}}{2\pi} e^{i \sum_{\mu} \bar{h}_{i_1}^{\mu} h_{i_1}^{\mu} - \frac{1}{2\lambda_1 N_1} \left[\sum_{\mu} s^{\mu} \sigma(h_{i_1}^{\mu}) \right]^2} - e^{-\frac{1}{2\lambda_0 N_0} \sum_{i_0} \left(\sum_{\mu} \bar{h}_{i_1}^{\mu} x_{i_0}^{\mu} \right)^2} \right\}^{N_1}, \quad (38)$$

where we used the fact that the integrals on $h_{i_1}^{\mu}$ and $\bar{h}_{i_1}^{\mu}$ can be factorized on the index i_1 . The integral over the $\bar{h}^{\mu}$ is Gaussian and can be solved:

$$\int \prod_{\mu}^P \frac{d\bar{h}^{\mu}}{2\pi} e^{i \sum_{\mu} \bar{h}^{\mu} h^{\mu} - \frac{1}{2\lambda_0 N_0} \sum_{i_0} \left(\sum_{\mu} \bar{h}^{\mu} x_{i_0}^{\mu} \right)^2} = P_1(\{h^{\mu}\}), \quad (39)$$

where

$$P_1(\{h^{\mu}\}) = \frac{e^{-\frac{1}{2} \sum_{\mu, \nu} h^{\mu} C_{\mu, \nu}^{-1} h^{\nu}}}{\sqrt{(2\pi)^P \det C}}, \quad C_{\mu, \nu} = \frac{1}{\lambda_0 N_0} \sum_{i_0} x_{i_0}^{\mu} x_{i_0}^{\nu}. \quad (40)$$

This last step requires the covariance matrix C to be invertible. Note that this is false as soon as $P > N_0$, but adding a small diagonal term to C solves the issue. One can explicitly check that the final result does not depend on this extra regularization.

To deal with the integral over h^{μ} we can include a further Dirac delta identity for the random variable $q = 1/\sqrt{\lambda_1 N_1} \sum_{\mu} \bar{s}^{\mu} \sigma(h^{\mu})$. This leaves us with the problem of finding the probability density $P(q)$. In the limit defined in equation (1), this is exactly the same setting of the Breuer–Major theorems^{44,45}, and, in particular, of the extension⁴³. As such, it is sufficient that both the (regularized) covariance C and the activation function σ satisfy the hypotheses of the theorem to guarantee that the probability distribution $P(q)$ converges in distribution to a Gaussian:

$$P(q) = \int d^P h P_1(\{h^{\mu}\}) \delta \left[q - \frac{1}{\sqrt{\lambda_1 N_1}} \sum_{\mu} \bar{s}^{\mu} \sigma(h^{\mu}) \right], \quad (41)$$

$$P(q) \rightarrow \mathcal{N}_q(0, Q).$$

with variance

$$Q(\bar{s}, C) = \frac{1}{\lambda_1 N_1} \sum_{\mu, \nu}^P \bar{s}^{\mu} \left[\int d^P h P_1(\{h^{\mu}\}) \sigma(h^{\mu}) \sigma(h^{\nu}) \right] \bar{s}^{\nu} = \frac{1}{\lambda_1 N_1} \sum_{\mu, \nu}^P \bar{s}^{\mu} K_{\mu, \nu}(C) \bar{s}^{\nu}. \quad (42)$$

One can show that there exist special configurations $\bar{s}$ in the domain of integration for which we are not allowed to invoke a Gaussian equivalence (see, for instance, our discussion at the end of Supplementary Section 2). In our derivation, we are assuming that the contribution of these special configurations to the effective action is negligible in the thermodynamic limit. Here we have also assumed that the variable q has zero mean, a condition verified as long as

$$\int d^P h P_1(\{h^{\mu}\}) \sigma(h^{\nu}) = 0, \quad (43)$$

that is, whenever σ is zero-mean; see Supplementary Section 4 for a more general derivation that is relevant for finite-mean activation functions such as ReLU.

Each element of the kernel matrix $K_{\mu, \nu}(C)$ can be easily reduced from a P -dimensional integral to a simpler two-dimensional one:

$$K_{\mu, \nu}(C) = \int \frac{dt_1 dt_2}{\sqrt{(2\pi)^2 \det \tilde{C}}} e^{-\frac{1}{2} \mathbf{t}^T \tilde{C}^{-1} \mathbf{t}} \sigma(t_1) \sigma(t_2), \quad (44)$$

where $\mathbf{t} = (t_1, t_2)^T$ and

$$\tilde{C} = \begin{pmatrix} C_{\mu\mu} & C_{\mu\nu} \\ C_{\mu\nu} & C_{\nu\nu} \end{pmatrix}. \quad (45)$$

is the reduced 2×2 input covariance matrix. It is worth pointing out that the kernel we find here is the so-called NNGP kernel. It differs from the neural tangent kernel that is found in the infinite-width limit of networks trained under gradient descent⁷⁰. The fact that the infinite-width limit of a Bayesian neural network differs from the one obtained from gradient descent is indeed known and discussed in literature²⁰.

Now we can integrate over the variable q and obtain:

$$\left[\int \frac{dq e^{-\frac{q^2}{2} - \frac{q^2}{2Q(\bar{s}, C)}}}{\sqrt{2\pi Q(\bar{s}, C)}} \right]^{\frac{N_1}{2}} = [Q(\bar{s}, C) + 1]^{-\frac{N_1}{2}}. \quad (46)$$

In the general case of finite $\alpha_1 = P/N_1$, we are only left with the integrals in s^{μ} and $\bar{s}^{\mu}$. To solve them it is convenient to introduce one final Dirac delta identity:

$$1 = \int dQ \delta \left[Q - \frac{1}{\lambda_1 N_1} \sum_{\mu, \nu} \bar{s}^{\mu} K(C)_{\mu, \nu} \bar{s}^{\nu} \right], \quad (47)$$

where $Q \geq -1$ is now an integration variable and not a function of $\bar{s}$, so that we have removed the explicit dependence on $\sqrt{Q(\bar{s}, C) + 1}$ in the partition function. Finally, the integrals in s^{μ} and $\bar{s}^{\mu}$ are Gaussian once another integral representation of the delta via a conjugate variable $\bar{Q}$ is inserted. This allows us to get the final effective action obtained in equation (7).

Exact solution of the saddle-point equations in the zero-temperature limit

The saddle-point equations obtained from equation (7) considerably simplify in the zero-temperature limit ($\beta \rightarrow \infty$). In particular, using the fact that the kernel K has only positive eigenvalues (in the asymptotic

regime α_1, α_0 finite), we get:

$$\bar{Q} = \frac{1}{1+Q}, Q = \frac{\alpha_1}{Q} - \frac{\alpha_1}{Q^2} y^T \left(\frac{K}{\lambda_1} \right)^{-1} y. \quad (48)$$

Given the condition $Q \geq -1$, the unique exact solution for $\bar{Q}$ is positive and reads:

$$\bar{Q}^* = \frac{\sqrt{(\alpha_1 - 1)^2 + 4\alpha_1 \frac{1}{P} y^T \left(\frac{K}{\lambda_1} \right)^{-1} y} - (\alpha_1 - 1)}{2}. \quad (49)$$

Predictors statistics

The main observable we are interested in is the generalization error (equation (5)). We can proceed along the same lines of the calculation performed in the previous section, introducing—other than the variables s^μ, h^μ defined by equations (36) and (37)—further variables s^0, h^0 that describe output and pre-activations of the new test example. We thus get:

$$\begin{aligned} \langle \epsilon_g(\mathbf{x}^0, y^0) \rangle &= \frac{1}{Z} \int \frac{ds^0 d\bar{s}^0}{2\pi} \int \prod_{\mu=1}^P \frac{ds^\mu d\bar{s}^\mu}{2\pi} (y^0 - s^0)^2 \\ &\times e^{-\frac{\beta}{2} \sum_{\mu=1}^P (y^\mu - s^\mu)^2 + i \sum_{\mu=1}^P s^\mu \bar{s}^\mu + i s^0 \bar{s}^0} \\ &\times \left[1 + \frac{1}{\lambda_1 N_1} \left(\sum_{\mu, \nu=1}^P \bar{s}^\mu K_{\mu\nu} s^\nu + 2s^0 \sum_{\mu=1}^P \bar{s}^\mu K_{\mu 0} (\mathbf{x}^0) + (s^0)^2 \kappa_0(\mathbf{x}^0) \right) \right]^{-\frac{N_1}{2}}, \end{aligned} \quad (50)$$

where κ_μ and κ_0 are the train–test and the test–test kernel integrals, respectively, as defined in equation (42) when the covariance matrix involves the test input, namely:

$$\begin{aligned} \kappa_\mu &= \int \frac{dt_1 dt_2}{\sqrt{(2\pi)^2 \det \tilde{C}_\mu}} e^{-\frac{1}{2} t^T \tilde{C}_\mu^{-1} t} \sigma(t_1) \sigma(t_2), \\ \kappa_0 &= \int \frac{dt}{\sqrt{2\pi C_{00}}} e^{-\frac{t^2}{2C_{00}}} \sigma(t)^2, \end{aligned} \quad (51)$$

where

$$\begin{aligned} \tilde{C}_\mu &= \begin{pmatrix} C_{\mu\mu} & C_{\mu 0} \\ C_{\mu 0} & C_{00} \end{pmatrix}, C_{\mu 0} = \frac{1}{\lambda_0 N_0} \sum_{i_0}^{N_0} x_{i_0}^\mu x_{i_0}^0, \\ C_{00} &= \frac{1}{\lambda_0 N_0} \sum_{i_0}^{N_0} (x_{i_0}^0)^2. \end{aligned} \quad (52)$$

Now we can introduce the order parameters Q and $\bar{Q}$ via equation (47) and their Fourier representation and perform the integration over all $s^\mu, \bar{s}^\mu$, and over s^0 . Doing so yields a single integral in s^0 and integrals on Q and $\bar{Q}$.

$$\langle \epsilon_g(\mathbf{x}^0, y^0) \rangle = \frac{1}{Z} \int \frac{dQ d\bar{Q}}{2\pi} e^{-\frac{N_1}{2} S(Q, \bar{Q})} \times \int \frac{ds^0 (y^0 - s^0)^2}{\sqrt{2\pi\sigma^2}} e^{-\frac{(s^0 + t_1)^2}{2\sigma^2}}, \quad (53)$$

with

$$\begin{aligned} \Gamma_1 &= \frac{\bar{Q}}{\lambda_1} \sum_{\mu\nu} K_{\mu\nu}(\mathbf{x}^0) \left(\frac{1}{\beta} + \frac{\bar{Q}}{\lambda_1} K \right)_{\mu\nu}^{-1} y_\nu, \\ \sigma_1^2 &= \frac{\bar{Q}}{\lambda_1} \left[\kappa_0(\mathbf{x}^0) - \frac{\bar{Q}}{\lambda_1} \sum_{\mu\nu} K_{\mu\nu}(\mathbf{x}^0) \left(\frac{1}{\beta} + \frac{\bar{Q}}{\lambda_1} K \right)_{\mu\nu}^{-1} \kappa_\nu(\mathbf{x}^0) \right] \end{aligned} \quad (54)$$

We can then unfold the easy integrals in s^0 and evaluate the result on the saddle-point solution. The generalization error is expressed

in terms of Γ_1 and σ_1^2 as in equation (8). Taking the $\beta \rightarrow \infty$ limit in equation (54) yields the expressions in equation (8).

Constraints on the scaling of the size of the dataset P with the input dimension N_0

In this section we address the additional constraints to the thermodynamic scaling ($P, N_1 \rightarrow \infty$ with $\alpha_1 = P/N_1$ finite) that may come from the hypotheses of the Breuer–Major on the covariance matrix C . The only stringent condition to verify is equation (29), that is

$$\sum_{\mu=1}^P |C_{\mu\nu}|^R < B_0 \quad \forall \nu = 1, \dots, P, \quad (55)$$

where B_0 is a given finite constant and R the Hermite rank of the activation function σ . In the case of inputs $\mathbf{x}$ with independent and identically distributed standard Gaussian coordinates, $C_{\mu\nu}$ is a Wishart random matrix with off-diagonal entries of order $1/\sqrt{N_0}$ and random signs: after taking the absolute value, the sum in equation (55) is of order $P(N_0)^{-R/2}$. Note that this provides an infinite class of activation functions (those with Hermite rank $R \geq 2$) where we can safely work at least at finite $\alpha_0 = P/N_0$. For activation functions with Hermite rank $R = 1$ (such as Erf or ReLU) we cannot provide such a guarantee by only looking at the hypothesis of the Breuer–Major theorem. It is also worth stressing that, given any odd (non-odd) activation function $\sigma(x)$ with Hermite rank $R = 1$, it is easy to engineer a new reasonable activation function with Hermite rank $R = 3$ ($R = 2$), just by replacing the old activation function with a new one $\sigma_1(x) = \sigma(x) - g_1 x$, where the coefficient $g_1 = \langle \sigma(x) \text{He}_1(x) \rangle$ and the average is over a normal distribution of zero mean and unit variance.

We observe that there is at least one case of activation function with $R = 1$ where the derivation goes through at finite α_0 , that is the linear function $\sigma(x) = x$ (in this case we can obtain the result at finite P, N_1, N_0 , as done also in ref. 55). In Supplementary Section 2 we examine the specific case of quadratic activation $\sigma(x) = x + x^2$ (which has $R = 1$), deriving the final effective action without employing the Breuer–Major theorem. As in the linear case, this derivation goes through at finite α_0 . We are thus led to believe that the scaling $P = O(\sqrt{N_0})$ suggested for $R = 1$ is overly pessimistic.

Generalization to deep neural networks with a finite number of hidden layers $L > 1$ and zero-mean activation

In the same spirit of the IHL calculation, we introduce L sets of auxiliary variables $h_{i_\ell}^\mu$ (where $i_\ell = 1, \dots, N_\ell$) that are equal to the pre-activations at each layer. The strategy to perform the calculation is to show that the probability distribution of the pre-activations at each layer $P_\ell(\{h_{i_\ell}^\mu\})$ can be computed recursively, starting from the input layer. We notice that this is conceptually different from the backpropagating kernel renormalization group introduced in ref. 5. It is still a kernel renormalization group, but forward-propagating and represents a generalization to NNTPs of the kernel recurrence arising in NNGPs¹⁵. In practice, our approach amounts to a systematic, layer-by-layer description of the pre-activation statistics by the Student's t -distribution that is shown to appear in the IHL case. This can be seen as a quantitative correction to the standard Gaussian statistics that is recovered in the infinite-width limit. At the moment we are not able to rederive the same result using the backpropagating method introduced in ref. 5.

Let us start by integrating the weights of the first layer. This defines a probability distribution over the pre-activations of the first layer via:

$$\begin{aligned} P_1(\{h_{i_1}^\mu\}) &= \int \mathcal{D}W^{(1)} \prod_{i_1, \mu} \delta \left(h_{i_1}^\mu - \frac{1}{\sqrt{N_0}} \sum_{i_0=1}^{N_0} W_{i_1 i_0}^{(1)} x_{i_0}^\mu \right) \\ &= \prod_{i_1=1}^{N_1} \frac{e^{-\frac{1}{2} \sum_{\mu=1}^M h_{i_1}^\mu C_{\mu\nu}^{-1} h_{i_1}^\nu}}{\sqrt{(2\pi)^P \det C}}. \end{aligned} \quad (56)$$

where C is defined in equation (40). This result is straightforward and it is valid for any N_0 , P and N_l , as the prior for the weights is Gaussian. At the second layer we have:

$$P_2(\{h_{i_2}^\mu\}) = \int \mathcal{D}W^{(2)} \mathcal{D}h_1 P_1(\{h_{i_1}^\mu\}) \times \prod_{i_2, \mu} \delta\left(h_{i_2}^\mu - \frac{1}{\sqrt{N_1}} \sum_{i_1=1}^{N_1} W_{i_2 i_1}^{(2)} \sigma(h_{i_1}^\mu)\right) \quad (57)$$

We now introduce conjugate variables $\tilde{h}_{i_2}^\mu$ to the activation of the second layer and the calculation proceeds as in the case of IHL architectures. To make analytical progress we need to make two fundamental approximations: (1) we assume that the set of random variables $q_{i_2} = 1/(\sqrt{N_1} \lambda_1) \sum_{i_1} \tilde{h}_{i_2}^\mu \sigma(h_{i_1}^\mu)$, where $\mathbf{h} \approx \mathcal{N}(0, C)$, is Gaussian-distributed; (2) we neglect correlations between different pre-activations of the second hidden layer. In conclusion, we get:

$$P_2(\{h_{i_2}^\mu\}) = \int dQ_1 d\bar{Q}_1 e^{-\frac{N_1}{2}(-Q_1 \bar{Q}_1 + \log(1+Q_1))} \times \prod_{i_2=1}^{N_2} \frac{e^{-\frac{1}{2} \sum_{\mu} \tilde{h}_{i_2}^\mu (Q_1 K(C)/\lambda_1)^{-1} \tilde{h}_{i_2}^\mu}}{\sqrt{(2\pi)^P \det(Q_1 K(C)/\lambda_1)}}, \quad (58)$$

where $K(C)$ is defined by equation (42). Notice that except for the integration over the two variables Q_1 and $\bar{Q}_1$, this is the same as the probability distribution of the IHL system (equation (40)) if we replace C with $\bar{Q}_1 K(C)/\lambda_1$. This reasoning can be easily iterated across layers and gives:

$$P_L(\{h_{i_L}^\mu\}) = \int \prod_{\ell=1}^{L-1} dQ_\ell d\bar{Q}_\ell e^{-\frac{N_\ell}{2}[-Q_\ell \bar{Q}_\ell + \log(1+Q_\ell)]} \times \prod_{i_L=1}^{N_L} \frac{e^{-\frac{1}{2} \sum_{\mu} \tilde{h}_{i_L}^\mu (K^{(R)}_{L-1}(\{Q_\ell\})^{-1} \tilde{h}_{i_L}^\mu}}{\sqrt{(2\pi)^P \det(K^{(R)}_{L-1}(\{Q_\ell\}))}}, \quad (59)$$

where $K^{(R)}_\ell(\{\bar{Q}_\ell\})$ is a renormalized kernel that satisfies the recurrence relation in equation (15).

The computation of the generalization error over a new example $(\mathbf{x}^0, y^0)$ gives:

$$\langle \epsilon_g(\mathbf{x}^0, y^0) \rangle = (y^0 - \Gamma_L)^2 + \sigma_L^2 \quad (60)$$

where Γ_L and σ_L^2 are defined in equation (17). Note that $K^{(R)}_{L\mu}$, $K^{(R)}_{L0}$ are recursive kernels that generalize the train–test and test–test kernels (equation (51)). They are defined starting from equation (15) where the kernel K is now evaluated with the covariance matrix C involving train–test or test–test points. Note that L -hidden layers generalization error is found replacing the IHL kernel with its recursive generalization (equation (15)).

Numerical experiments

Network architectures. We perform numerical experiments with deep fully connected architectures trained on two regression tasks in computer vision. In particular we use the 0 and 1 classes of the MNIST and CIFAR10 datasets, which, for the latter, correspond to the labels ‘cars’ and ‘planes’ (Fig. 1a). Examples from CIFAR10 are coarse grained to $N_0 = 28 \times 28$ pixels and converted to greyscale.

To test our theory in the zero-mean activation function case, we used the Erf function, for which the NNGP kernel can be computed analytically⁷¹:

$$K_{\mu\nu}^{\text{Erf}}(C) = \frac{2}{\pi} \arcsin\left(\frac{2C_{\mu\nu}}{\sqrt{(1+2C_{\mu\mu})(1+2C_{\nu\nu})}}\right). \quad (61)$$

In Fig. 2 we train networks with $\sigma = \text{ReLU}$. The kernel can be computed analytically also in this case⁶³ and reads:

$$K_{\mu\nu}^{\text{ReLU}}(C) = \sqrt{C_{\mu\mu} C_{\nu\nu}} \kappa\left(\frac{C_{\mu\nu}}{\sqrt{C_{\mu\mu} C_{\nu\nu}}}\right), \quad (62)$$

$$\kappa(x) = \frac{1}{2\pi} \left[x(\pi - \arccos(x)) + \sqrt{1-x^2} \right].$$

Sampling from the Bayesian posterior. To ensure convergence of the posterior weights distribution to the Gibbs ensemble, we train our networks using a discretized Langevin dynamics, similarly to what is done in refs. 3,5. At each training step t , the parameters $\theta = \{W^\ell, v\}$ are updated according to:

$$\theta(t+1) = \theta(t) - \eta \nabla_\theta \mathcal{L}(\theta(t)) + \sqrt{2\eta} \epsilon(t) \quad (63)$$

where $T = 1/\beta$ is the temperature, η is the learning rate, $\epsilon(t)$ is a white Gaussian noise vector with entries drawn from a standard normal distribution, and the loss is the one defined in equation (3). We employ $T = \eta = 10^{-3}$ throughout all the experiments. This is sufficient to approximate the $T = 0$ dynamics in the regime we are considering. This dynamics requires $10^5/10^6$ steps to reach thermalization, depending on the sizes of the dataset and network. We extract the generalization loss within a single run: after the train error has reached its minimum and the test loss is thermalized, we average test loss values every $10^3/10^4$ epochs (depending again on the magnitude of P , N_ℓ). For the sake of completeness, we report the best test accuracy achieved on both datasets by IHL architectures: 0.86 on CIFAR10 with $P = 3,000$ and $\lambda_1 = 1,000$, 0.999 on MNIST with the same Gaussian prior and $P = 1,000$. The train accuracy is always 1. Further comments on the technical issues encountered in simulating the Bayesian dynamics are discussed in Supplementary Section 5.

Data availability

The CIFAR10⁷² and MNIST⁷³ datasets that we used for all our experiments are publicly available online, respectively, at <https://www.cs.toronto.edu/~kriz/cifar.html> and <http://yann.lecun.com/exdb/mnist/>.

Code availability

The code used to perform experiments, compute theory predictions and analyse data is available at: https://github.com/rpacelli/FC_deep_bayesian_networks (ref. 74).

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Author contributions

P.R., S.A. and M.P. performed the analytical calculations, supported by F.G., M.G. and R.P. Numerical experiments, data analysis and data visualization were carried out by R.P. All the authors contributed to discussing and interpreting the results and to writing and editing the paper. S.A. and R.P. contributed equally to the work.

Competing interests

The authors declare no competing interests.

Additional information

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